

Marco Tamburini

Tampa, FL • (813) 399-3794 • marcotamburini@usf.edu • [linkedin.com/in/marco-tamburini31](https://www.linkedin.com/in/marco-tamburini31) • <https://github.com/MarcoTamburini> • <https://engineering-portfolio-steel.vercel.app/>

EDUCATION

University of South Florida
Bachelor of Science in Electrical Engineering | GPA: 4.00/4.00

Tampa, FL
Expected May 2029

EXPERIENCE

Newkirk Electric Associates

Muskegon, MI

Power Engineering Intern

Summer 2026

- Supported prefabricated material racking and distribution for a large-scale **data center power generation project**, coordinating between the Senior Engineering Manager and CTS Manager to maintain schedule and material accuracy.
- Developed **raceway and cable support sizing and layout documentation**, translating field requirements into structured technical deliverables for project managers and field crews.

PROJECTS

Wireless Power Transfer (WPT) System Simulator

Summer 2026 | MATLAB | LTspice | Python

- Architected a dual-regime WPT link budget simulator in **MATLAB App Designer**, modeling **far-field Friis transmission** and **near-field inductive coupling** (Neumann mutual inductance, Wheeler spiral formula) in a unified interactive GUI.
- Optimized RF front-end performance via S11 parameter analysis and simulated impedance matching networks for power transfer efficiency.
- Implemented a **frequency-hysteresis state machine** for regime auto-switching, a coupled-resonator efficiency engine, load resistance optimizer, and an **HSMS-2850 heuristic rectifier model** for full-stack DC output estimation.

SYP – Startup Venture | Hardware Engineering Contributor

2026 – Present | RFID

- Engineered a **patent-pending** RFID-based athlete identification system for **D1 athletics**; presented integrated hardware to RDM Innovation and won **1st place** at the USF Innovation Fest Solve the Startup Challenge (March 2026).
- Selected as **1 of 25 national finalists** from 170+ submissions (65+ universities) at the Schulze Entrepreneurship Challenge (e-Fest, Minneapolis, April 2026), securing **\$10,000** in prize funding.

Buck Converter – 5A Power Supply

Spring 2026 | Altium Designer | PSpice | LTspice

- Designed a **5A synchronous buck converter** (7S3P Li-ion, 24–29V to 5V) for the USF Electrathon vehicle; created full schematic and multi-layer PCB in Altium Designer and achieved **Altium Certification**; validated waveforms and efficiency in PSpice/LTspice prior to fabrication.
- Optimized SI/PI and thermal dissipation via strategic trace routing, polygon pours, and thermal vias, mitigating voltage drops under high-current loads.

VBT Device – Velocity-Based Training Tool

Winter 2026 | IMU | SolidWorks | BLE | C++ | Flutter

- Designed an IMU-based device utilizing the Madgwick sensor fusion algorithm to measure barbell and cable-machine velocity and ROM in real time, validated to **within 5% accuracy** across 10+ movement patterns.
- Developed custom C++ BLE firmware and a companion Flutter mobile app with Firebase backend to stream and log real-time kinematic data.

LEADERSHIP & INVOLVEMENT

IEEE USF Chapter | Professional Development Chair – E-Board

April 2026 – Present

- Organize industry speaker sessions and career events for 400+ engineering students; maintain a live REU repository on the chapter website; launched a cross-club Discord network across USF Engineering.
- Partner with the Nault Center for Entrepreneurship, USF Connect, and the USF Patent Office to develop student innovation and IP commercialization programming.

USF College of Engineering | Engineering Ambassador – E-Board, Special Events Coordinator

Sept 2025 – Present

- Recruit and manage a cohort of 30+ ambassadors, overseeing interview and selection processes to maintain program quality.
- Coordinated major outreach events**, including EXPO and SHPE Jr. Noche de Ciencias, while representing the College during prospective student tours to drive recruitment.

TECHNICAL SKILLS

Circuit Design & Simulation: Altium Designer (Certified), PSpice, LTspice, Simulink

Embedded & Firmware: C++, BLE, RFID, IMU integration, microcontroller programming, Flutter (Dart), Firebase

Software & Tools: MATLAB, Git, Fusion 360, Python, n8n